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THE PSEUDORANDOM SEQUENCE OF ARRAYS*

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Abstract. Pseudorandom (or maximal length) sequences and arrays have been known for a long time, and have been reported by several authors. Pseudorandom volumes have also been mentioned.

This paper presents a different type of sequence of arrays in which the arrays share many properties with the pseudorandom arrays of the literature, and the sequence (or group) of arrays has many properties in common with the pseudorandom sequences. It is proposed that this set of arrays be called a "pseudorandom sequence of arrays," or PRSAs. Some interesting properties of the PRSA as well as its practical (hardware) implementation have been mentioned. It has also been shown that our result is a special case of the general $N(3)/D(3)$ case.

1. Introduction

Shift register generated pseudorandom sequences are familiar and have been reported by several authors [1]–[5] in various contexts. It is well known [6] that an m-bit shift register generator (SRG) can produce at most m linearly independent sequences. A criterion to determine linear independence among distinct sequences from an SRG has been given in [7] and it has been shown in [8] that if m linearly independent sequences forming an SRG are placed side by side, all possible nonzero m-tuples will be exhausted. The window property of pseudorandom sequences [2] guarantees that if a window of width m is slid over a pseudorandom sequence for an m-bit SRG, then each m-tuple is seen once and only once during the window sliding operation.

It is shown in this paper that the exhaustion of possibilities in the two directions can be combined to get what we call a pseudorandom sequence of arrays (PRSA). Some interesting cases of the PRSA are mentioned

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followed by a few properties of the PRSA. A hardware implementation for the PRSA generator is also given.

2. Construction of the PRSA

Pick m linearly independent pseudorandom sequences, each having a length of $n = 2^m - 1$, from an m-bit SRG and form an $m \times n$ matrix L. Each sequence forms a row of L. All the rows of L can be generated simultaneously from an m-bit SRG by using a configuration slightly different from the basic configuration of the SRG [9], which we call the modified SRG with m outputs and is shown in Figure 1.

The PRSA Theorem. *Imagine that the $m \times n$ matrix L is spread on the surface of a cylinder. If an $m \times m$ window is slid over L, then every $m \times m$ submatrix is seen only once.*

Proof. It has been shown by [8] that every column of L is distinct because the rows of L are linearly independent pseudorandom sequences. So each shift of the $m \times m$ window encloses m distinct columns of L. Also since each row of L is a pseudorandom sequence, each m-bit row seen during each shift of the $m \times m$ window is distinct by the window property [2] of pseudorandom sequences. Hence every $m \times m$ submatrix is seen only once. □

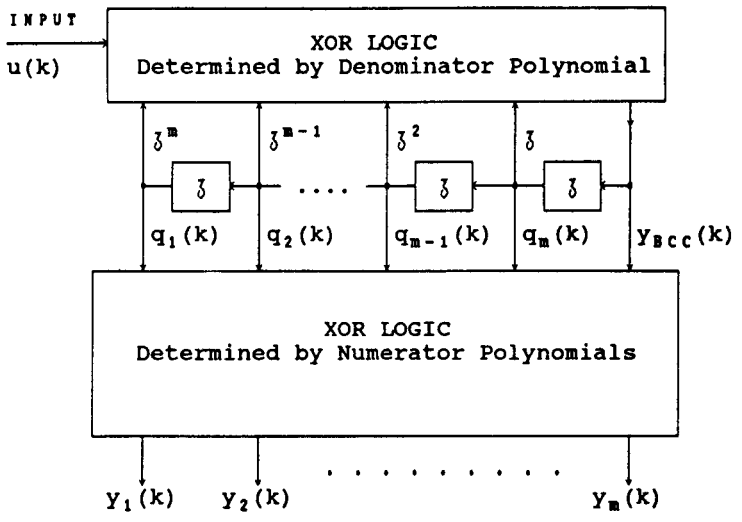


Figure 1. A modified SRG with m outputs.

Define $\{X(k): 0 \le k \le n-1\} = \{X(0), X(1), \ldots, X(n-1)\}$ to be the sequence of $m \times m$ submatrices seen during the shifting process as mentioned in the above theorem. We call this sequence a PRSA because it is in many ways similar to the pseudorandom sequences, while the arrays forming the elements of this sequence are in many ways similar to the pseudorandom arrays. Some properties of the PRSA are listed and proved in the following section, and the similarities and differences between the PRSA, pseudorandom sequences, and pseudorandom arrays are also mentioned.

3. Generating the PRSA

In terms of hardware, the PRSA can easily be generated by connecting m outrigger shift registers (OSRs) to the "modified SRG with m outputs" of Figure 1. The elements of a particular $m \times m$ array, $X(k)$, are taken at the outputs of the delayors, as shown in Figure 2. Each clock pulse produces the next array of the PRSA at these taps. Figure 3 shows the complete PRSA generator for an example with

$$N_1(3) = 1, \qquad N_2(3) = 1 \oplus 3, \qquad N_3(3) = 1 \oplus 3 \oplus 3^2,$$
$$N_4(3) = 1 \oplus 3 \oplus 3^3, \qquad N_5(3) = 1 \oplus 3 \oplus 3^4,$$

and

$$P(3) = 1 \oplus 3^3 \oplus 3^5.$$

The PRSA itself is listed in Table 1.

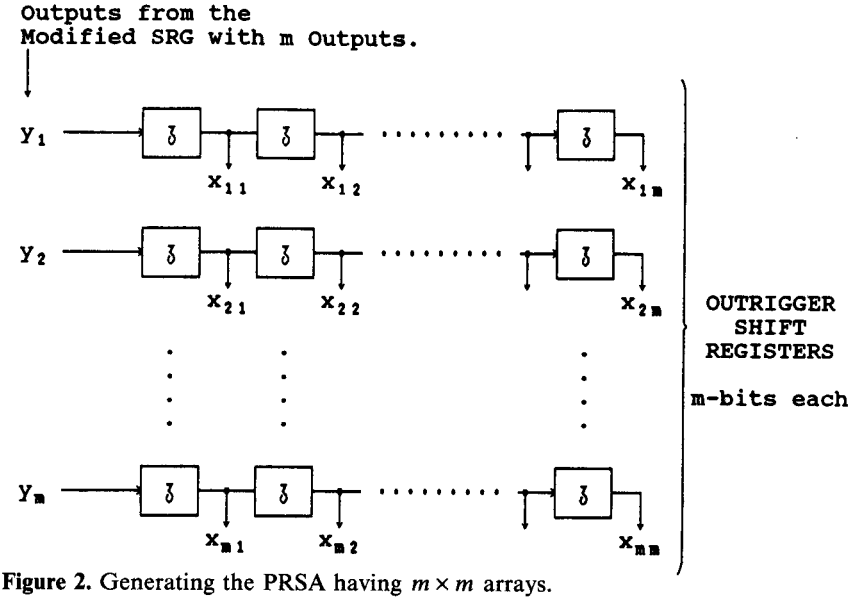


Figure 2. Generating the PRSA having $m \times m$ arrays.

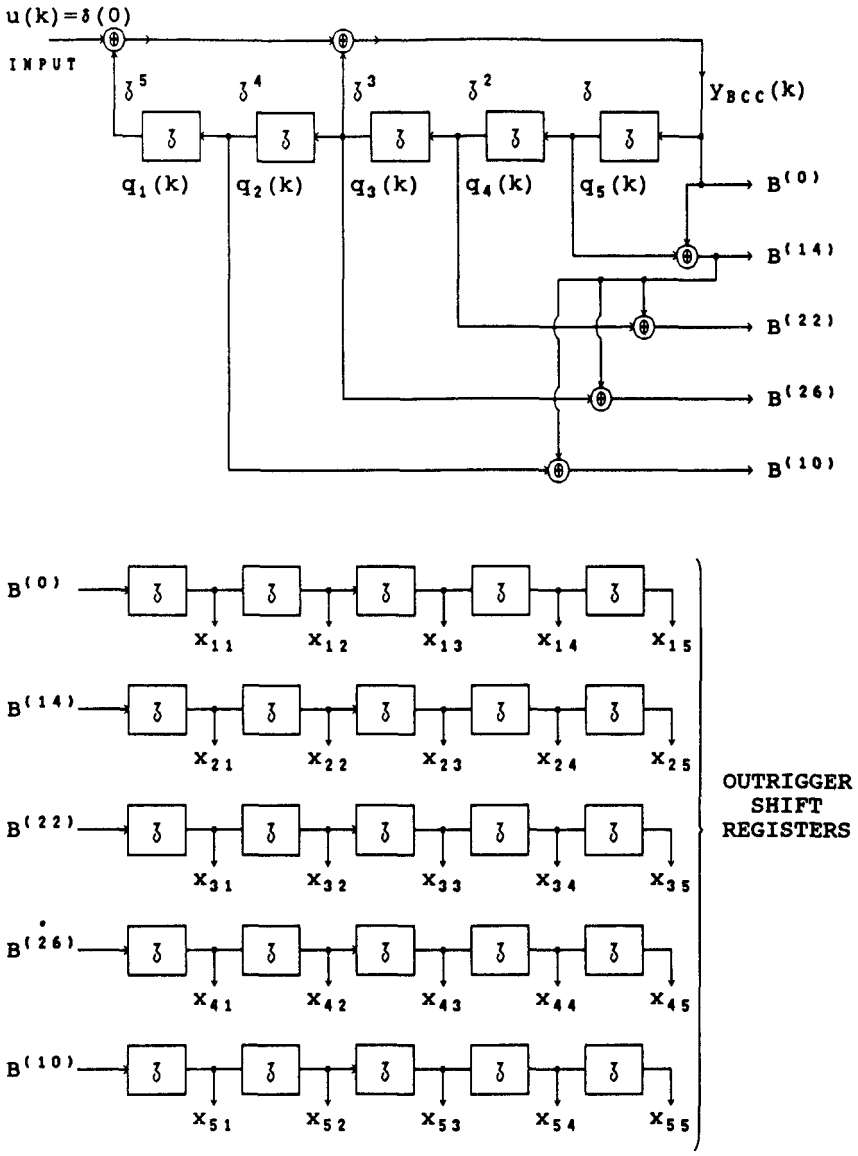


Figure 3. A complete PRSA generator for $m = 5$.

Table 1. One period from a 5-bit type 1 PRSA generator. (Characteristic polynomial: $P(3) = 1 \oplus 3^3 \oplus 3^5$. Numerator polynomials: $N_1(3) = 1$, $N_2(3) = 1 \oplus 3$, $N_3(3) = 1 \oplus 3 \oplus 3^2$, $N_4(3) = 1 \oplus 3 \oplus 3^3$, $N_5(3) = 1 \oplus 3 \oplus 3^4$.)

k	$X(k)$	k	$X(k)$	k	$X(k)$	k	$X(k)$
0	0 1 0 0 1	1	1 0 1 0 0	2	1 1 0 1 0	3	0 1 1 0 1
	1 1 0 1 1		1 1 1 0 1		0 1 1 1 0		1 0 1 1 1
	1 1 1 1 1		0 1 1 1 1		0 0 1 1 1		0 0 0 1 1
	1 0 0 1 1		1 1 0 0 1		1 1 1 0 0		1 1 1 1 0
	0 1 0 1 1		1 0 1 0 1		0 1 0 1 0		0 0 1 0 1
4	0 0 1 1 0	5	1 0 0 1 1	6	1 1 0 0 1	7	1 1 1 0 0
	0 1 0 1 1		1 0 1 0 1		0 1 0 1 0		0 0 1 0 1
	1 0 0 0 1		1 1 0 0 0		0 1 1 0 0		1 0 1 1 0
	1 1 1 1 1		0 1 1 1 1		0 0 1 1 1		0 0 0 1 1
	0 0 0 1 0		0 0 0 0 1		1 0 0 0 0		0 1 0 0 0
8	1 1 1 1 0	9	1 1 1 1 1	10	0 1 1 1 1	11	0 0 1 1 1
	0 0 0 1 0		0 0 0 0 1		1 0 0 0 0		0 1 0 0 0
	1 1 0 1 1		1 1 1 0 1		0 1 1 1 0		1 0 1 1 1
	1 0 0 0 1		1 1 0 0 0		0 1 1 0 0		1 0 1 1 0
	0 0 1 0 0		1 0 0 1 0		0 1 0 0 1		1 0 1 0 0
12	0 0 0 1 1	13	1 0 0 0 1	14	1 1 0 0 0	15	0 1 1 0 0
	0 0 1 0 0		1 0 0 1 0		0 1 0 0 1		1 0 1 0 0
	0 1 0 1 1		1 0 1 0 1		0 1 0 1 0		0 0 1 0 1
	1 1 0 1 1		1 1 1 0 1		0 1 1 1 0		1 0 1 1 1
	1 1 0 1 0		0 1 1 0 1		0 0 1 1 0		1 0 0 1 1
16	1 0 1 1 0	17	1 1 0 1 1	18	1 1 1 0 1	19	0 1 1 1 0
	1 1 0 1 0		0 1 1 0 1		0 0 1 1 0		1 0 0 1 1
	0 0 0 1 0		0 0 0 0 1		1 0 0 0 0		0 1 0 0 0
	0 1 0 1 1		1 0 1 0 1		0 1 0 1 0		0 0 1 0 1
	1 1 0 0 1		1 1 1 0 0		1 1 1 1 0		1 1 1 1 1
20	1 0 1 1 1	21	0 1 0 1 1	22	1 0 1 0 1	23	0 1 0 1 0
	1 1 0 0 1		1 1 1 0 0		1 1 1 1 0		1 1 1 1 1
	0 0 1 0 0		1 0 0 1 0		0 1 0 0 1		1 0 1 0 0
	0 0 0 1 0		0 0 0 0 1		1 0 0 0 0		0 1 0 0 0
	0 1 1 1 1		0 0 1 1 1		0 0 0 1 1		1 0 0 0 1
24	0 0 1 0 1	25	0 0 0 1 0	26	0 0 0 0 1	27	1 0 0 0 0
	0 1 1 1 1		0 0 1 1 1		0 0 0 1 1		1 0 0 0 1
	1 1 0 1 0		0 1 1 0 1		0 0 1 1 0		1 0 0 1 1
	0 0 1 0 0		1 0 0 1 0		0 1 0 0 1		1 0 1 0 0
	1 1 0 0 0		0 1 1 0 0		1 0 1 1 0		1 1 0 1 1

(*continued*)

Table 1 (*continued*)

k	$X(k)$	k	$X(k)$	k	$X(k)$	k	$X(k)$
28	0 1 0 0 0	29	0 0 1 0 0	30	1 0 0 1 0		
	1 1 0 0 0		0 1 1 0 0		1 0 1 1 0		
	1 1 0 0 1		1 1 1 0 0		1 1 1 1 0		
	1 1 0 1 0		0 1 1 0 1		0 0 1 1 0		
	1 1 1 0 1		0 1 1 1 0		1 0 1 1 1		

4. Properties of the PRSA

PRSA 1 (Maximum Number of Elements). *The maximum number of elements that a PRSA can have is* $n = 2^m - 1$.

Proof. This is trivial because the PRSA is generated by the matrix L in which every row is a pseudorandom sequence and thus every row can have at most n distinct nonzero binary m-tuples seen during the "window sliding" operation. □

PRSA 2 (Pseudorandom Disparity). *The disparity (modulo-2 sum) of the elements of each* $X(k+1)$ *is either a* 1 *or a* 0, *independent of the disparity of* $X(k)$.

Proof. Again by the window property of pseudorandom sequences, each row of $X(k+1)$ is a distinct m-tuple of the generating pseudorandom sequence, independent of the corresponding row of $X(k)$. It may contain an even number of ones, in which case the elements of that row sum to a 0, or an odd number of ones, in which case the element sum of a 1. Extending this procedure to every row of $X(k+1)$ will result in a modulo-2 sum of all the elements of $X(k+1)$ to be either a 1 or a 0, independent of the sum of the elements of $X(k)$. □

PRSA 3 (The Shift Property).

(a) *The sequence* $\{X(k)\}$ *satisfies the shift property* [2] *of pseudorandom sequences.*

(b) *Each* $X(k)$ *itself satisfies the shift property* [2] *of pseudorandom arrays for horizontal shifts only.*

Proof. Trivial because of the way we have constructed the PRSA. □

PRSA 4 (Recurrence).

(a) *The sequence* $\{X(k)\}$ *satisfies the recurrence*

$$X(\kappa+m)=f_{m-1}X(\kappa+m-1)\oplus f_{m-2}X(\kappa+m-2)\oplus\cdots$$
$$\oplus f_1X(\kappa+1)\oplus X(\kappa),$$

where $\kappa=0,1,2,\ldots$ *and* $f_i=0$ *or* 1 *are the coefficients of the* 3^i *term in the characteristic polynomial* $P(3)$ *associated with the generating SRG. Conversely, any solution of the above equation is a maximal length PRSA associated with* $P(3)$*, and there are precisely* 2^m-1 *distinct solutions to the above equation. Out of these* 2^m-1 *solutions, only* m *are linearly independent. Thus there are at most* m *linearly independent maximal length PRSAs associated with* $P(3)$.

(b) *The rows and colmns of each* $X(k)$ *do not satisfy any recurrence.*

(c) *Each* $X(k+1)$ *is related to* $X(k)$ *by the following recurrence:*

$$X(k+1)=TX(k),$$

where T *is an* $m\times m$ *matrix determined (only) by the numerator and denominator polynomials in the PRSA generator.*

Proof. (a) Each element $x_{ij}(k)$ of $X(k)$ satisfies a recurrence

$$x_{ij}(\kappa+m)=f_{m-1}x_{ij}(\kappa+m-1)\oplus f_{m-2}x_{ij}(\kappa+m-2)\oplus\cdots$$
$$\times\oplus f_1x_{ij}(\kappa+1)\oplus x_{ij}(\kappa)$$

because it is an element of a pseudorandom sequence associated with $P(3)$. Thus the sequence $\{X(k)\}$ satisfies the same recurrence.

(b) Because of linear independence among the rows (and columns) of each $X(k)$, and the fact that each row (and column) of $X(k)$ is distinct, no recurrence exists along the vertical or horizontal directions within $X(k)$.

(c) With reference to Figure 2, if the jth state of the ith OSR is represented by $x_{ij}(k)$, then all the states can be written concisely as an $m\times m$ matrix $X(k)=[x_{ij}(k)]$. Moreover, since $y_i(k)=c_i^TQ(k)$ [10], we can form a vector $Y(k)$ whose elements are the $y_i(k)$'s, as shown below:

$$Y(k)=\begin{bmatrix}y_1(k)\\y_2(k)\\\vdots\\y_m(k)\end{bmatrix}=\begin{bmatrix}C_1^T\\C_2^T\\\vdots\\C_m^T\end{bmatrix}Q(k)=KQ(k) \tag{1}$$

and

$$Y(k+1)=KQ(k+1)=KAQ(k), \tag{2}$$

where A is the companion matrix of $P_R(3)$ corresponding to the generating SRG. See [10] for details. If K is invertible (a nonsingular transformation), then from (1) we get

$$Q(k) = K^{-1} Y(k). \tag{3}$$

Use (3) in (2)

$$Y(k+1) = (KAK^{-1}) Y(k) = TY(k), \tag{4}$$

where we have used $T = KAK^{-1}$, and T is determined completely by the numerator and denominator polynomials. Note that K^{-1} is calculated using modulo-2 arithmetic. Its existence is guaranteed because the elements of L are linearly independent.

It can be easily seen that the vector $Y(k)$ is a column of the matrix $X(k)$, so that we may write (for the periodic part)

$$X(k) = [\, Y(k+m-1) | \cdots | Y(k+1) | Y(k)]$$

which means that

$$\begin{aligned} X(k+1) &= [\, Y(k+m) | \cdots | Y(k+2) | Y(k+1)] \\ &= [\, TY(k+m-1) | \cdots | TY(k+1) | TY(k)] \\ &= TX(k). \end{aligned}$$

Thus every element, $X(k+1)$, of the PRSA is related to the previous element $X(k)$ by

$$\boxed{X(k+1) = KAK^{-1} X(k).} \qquad \square$$

PRSA 5 (The Window Property).

(a) *If the sequence* $\{X(0), X(1), \ldots, X(n-1)\}$ *is written on the surface of a cylinder and an* $m \times m^2$ *template is slid over it, then each set of* m *distinct arrays* (*of dimension* $m \times m$) *is seen only once. Figure* 4 *gives an example.*

(b) *The window property of pseudorandom arrays* [2] *is not satisfied in general.*

Proof. (a) Trivial because of the way we have constructed the PRSA.

(b) If m is prime, a window cannot be defined. If m is factorable, the window property is still not satisfied as shown by the following example.

Take the maximal length sequence from an SRG associated with

$$P(3) = 1 \oplus 3^3 \oplus 3^4$$

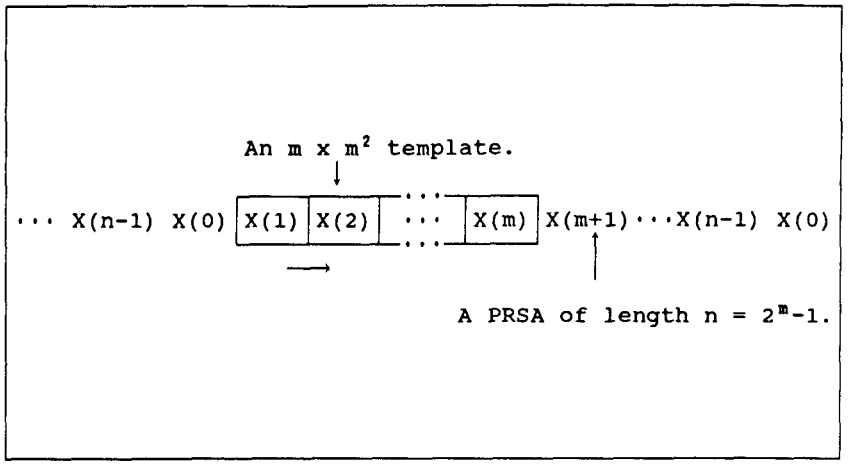


Figure 4. Explanation of the window property for a PRSA.

and form a PRSA by using

$$L=\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

so that

$$X(3)=\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$

Since $m=4$, the size of the window is 2×2 [2]. As shown below, the same 2×2 array is seen at two positions of the 2×2 window:

A 2×2 window

$$X(3)=\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$

□

Definition. If we consider the PRSA as a sequence $\{X(k)\}$ of period p, then the *autocorrelation function*, $C(\tau)$, can be defined [2] as follows. If A is the

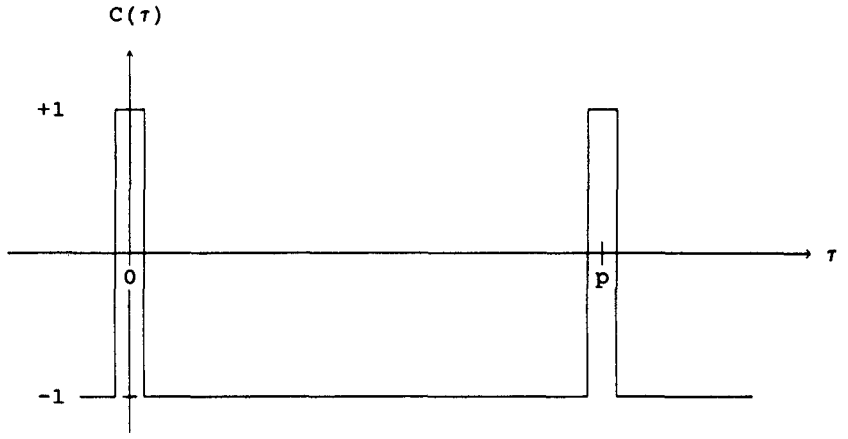


Figure 5. The periodic autocorrelation function of a PRSA.

number of places where $\{X(0), X(1), \ldots, X(p-1)\}$ and the cyclic shift $\{X(\tau), X(\tau+1), \ldots, X(\tau-1)\}$ agree, and D is the number of places where they disagree, then

$$C(\tau) = \frac{A-D}{A+D} = \frac{A-D}{p}; \qquad \tau = 0, \pm 1, \pm 2, \ldots.$$

Note that $p = n$ for a maximal length case.

PRSA 6 (Two-Valued Autocorrelation Function). *According to the above definition, the periodic autocorrelation function of every PRSA as a sequence of period p, is double valued, given by*

$$C(0) = +1 \quad \textit{for} \quad \tau = 0,$$

$$C(\tau) = -1 \quad \textit{for} \quad 1 \leq \tau \leq p-1.$$

This is plotted in Figure 5.

5. Discussion

A different type of sequence, the PRSA, has been presented and discussed in this paper. A few properties of the PRSA have been mentioned showing that the PRSA is in many ways similar to pseudorandom sequences and each member of the PRSA is in many ways similar to pseudorandom arrays. However, since each element of the PRSA is itself an array, a one-to-one correspondence between the PRSA and pseudorandom sequences does not exist. The PRSA does have a two-valued autocorrelation function, is periodic

in nature, and can have a maximum period of $n = 2^m - 1$. Thus only in some respects are PRSAs similar to pseudorandom sequences.

Each element of the PRSA is an $m \times m$ pseudorandom array in itself. So our PRSA is also a Sequence of Pseudorandom Arrays (SPRA). These arrays are square arrays with all possible dimensions, while the pseudorandom arrays reported in the literature are only almost-square and do not exist for all dimensions. The pseudorandom arrays of the literature are symmetrical about a center column of zeros, whereas the arrays mentioned in this paper do not possess any kind of symmetry. It is interesting to note that our PRSA generator is also capable of producing more than one subcyclic PRSA. This depends on the characteristic polynomial in addition to the "seed" ard thus does require a modification in hardware.

Our result can be thought of as a special case to the general $N(3)/D(3)$ case. By the general case, we mean that no restriction is imposed on the degree of $N(3)$. Thus whenever $\deg(N(3)) \geq \deg(D(3))$, we can separate the transfer function into two parts as explained below by taking an example.

Let

$$\frac{N(3)}{D(3)} = \frac{1 \oplus 3 \oplus 3^4 \oplus 3^7}{1 \oplus 3^3 \oplus 3^4} = \frac{1 \oplus 3}{1 \oplus 3^3 \oplus 3^4} \oplus \frac{3^4 \oplus 3^7}{1 \oplus 3^3 \oplus 3^4}.$$

The first term in the above expression can be handled in the usual way. For the second term, recall that since

$$D(3) = 0, \qquad 3^4 = 3^3 \oplus 1,$$

so

$$\begin{aligned} 3^7 &= 3^4 3^3 = 3^3(3^3 \oplus 1) = 3^6 \oplus 3^3 \\ &= 3^2(3^3 \oplus 1) \oplus 3^3 = 3^5 \oplus 3^3 \oplus 3^2 \\ &= 3(3^3 \oplus 1) \oplus 3^3 \oplus 3^2 = 3^4 \oplus 3^3 \oplus 3^2 \oplus 3 \\ &= 3^2 \oplus 3 \oplus 1. \end{aligned}$$

Thus we get

$$\begin{aligned} \frac{N(3)}{D(3)} &= \frac{(1 \oplus 3) \oplus (3^3 \oplus 1) \oplus (3^2 \oplus 3 \oplus 1)}{1 \oplus 3^3 \oplus 3^4} \\ &= \frac{1 \oplus 3^2 \oplus 3^3}{1 \oplus 3^3 \oplus 3^4} \end{aligned}$$

which is the same as our case. Thus whenever $\deg(N(3)) \geq \deg(D(3))$, it can be reduced to $\deg(N(3)) < \deg(D(3))$ by using the characteristic equation.

Figure 6 shows a pseudorandom array mentioned by Spann [11]. It is interesting to note that this construction corresponds to a "raster scan" array. Our PRSA generator can produce the "serpentine scan" array in

...	a_5	a_6	a_7	a_8	a_9	a_5	a_6	a_7	a_8	a_9
...	a_{10}	a_{11}	a_{12}	a_{13}	a_{14}	a_{10}	a_{11}	a_{12}	a_{13}	a_{14}
...	a_0	a_1	a_2	a_3	a_4	a_0	a_1	a_2	a_3	a_4
...	a_5	a_6	a_7	a_8	a_9	a_5	a_6	a_7	a_8	a_9
...	a_{10}	a_{11}	a_{12}	a_{13}	a_{14}	a_{10}	a_{11}	a_{12}	a_{13}	a_{14}
...	a_0	a_1	a_2	a_3	a_4	a_0	a_1	a_2	a_3	a_4
...	a_5	a_6	a_7	a_8	a_9	a_5	a_6	a_7	a_8	a_9
...	a_{10}	a_{11}	a_{12}	a_{13}	a_{14}	a_{10}	a_{11}	a_{12}	a_{13}	a_{14}
...	a_0	a_1	a_2	a_3	a_4	a_0	a_1	a_2	a_3	a_4

Figure 6. A 3×5 raster scan array.

addition to the raster scan array of [11]. This can be achieved by reversing the OSRs for even values of i in Figure 2.

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